

PROJECT HOOVER · T11 BABY HOOVER V4 · FIRST LIVE RUN

Heard nothing, learned plenty

Findings from the 5 October league-style run: the first time the story layer met a real field.

RUN	BRANCH / HEAD	SESSION	DATE
HOOVER_20261005_154004_s01	baby-hoover-v4-stories · 7220ee4 (as run)	14 min · two races · 20 cars · 2 humans	6 OCT 26 (run 5 OCT)

Executive summary

The broadcast was silent and the camera lived with the AI, but the machine underneath did what it was built to do: it ran live for fourteen minutes across two races, wrote 109 lines and every artefact, and the story layer processed 116 stories without a fault. The two things the room noticed were an old plumbing bug and a new camera rule. Both are fixed; so are the four script defects the data exposed.

Three symptoms, three different causes. **No speech:** ElevenLabs worked (five lines billed); playback failed five times with a `ValueError` because the audio device was addressed by name, which Windows lists under several host APIs, so the channel went to degraded mode and wrote the rest silently. This is Mike's 30 September failure; the patch lived on two machines and not in the branch. **Camera off the humans** (11% of hold time on your two cars): AI-only story moments, chiefly 38 collapse stories down the field, took the protected camera layer. **Model never wrote a line:** the hybrid writer's routing list named only the V3 kinds the stories now replace.

The script itself, read afterwards, was 59% one kind of line: 64 of 109 were the analyst relating midfield AI stories back to drivers who were running first and second ("All of that is fifteen places behind Valor"). True, and useless. The relate pass fired on proximity; it should fire on consequence.

109

lines written · 0 spoken · 109 failed (5 playback errors, then degraded)

11%

hold time on human cars (two humans of twenty)

64

of 109 lines were relate beats; analyst spoke 87 of 109

116

stories opened · 262 beats · no processor raised

What happened

Hoover joined a session already on its final lap (fallback anchor, 3:42), followed the finish, then picked up the restart at 3:45 with a proper lights-out and ran the second race to classification at 3:53. The V3 state machine handled the restart cleanly. Thirty-three processors registered, one dormant by design (yellow flags). The camera made 83 cuts, 50 protected, 38 of those for AI collapse stories.

HIGHLIGHTS

- **The chain ran live, end to end**, for a full session with the story layer on, and finalised correctly on the interrupt.
- **The stories were right.** A catch narrated as a catch: *"Pixeliell is closing on Valor... on the back of Valor... goes through on Valor for second."* Contact with its consequence: *"Contact! Valor and Pixeliell" → "Valor has lost two places out of that, down to fifth."* DRS calls, the human-order line, the cluster forming and splitting, "Two laps to go," the classification correction.
- **98% of story lines** carried a human subject or a relation to one. The Story-to-Human Relate mechanism works; its gate did not.
- **Keys and the bin were fine.** Both API keys were present; 110,804 packets recorded, replayable, and every fix overleaf was verified against them.

LOWLIGHTS

- **Silent broadcast.** Playback by device name, five failures, degraded mode. The failure reported as a count, not a cause; the cause was in the per-line records all along.
- **Camera on the AI.** Story moments outranked the human-default layer; collapses in the midfield held the shot.
- **Relate flood.** One line in every two was a relation to a driver the story could not touch.
- **Lead-battle churn.** The story was keyed on the chaser-leader pair; every time the car behind Ronin changed, the battle "ended" and "began" again, seconds apart.
- **Template-only script.** No model lines, so no context beyond rotation.
- **Phrasing faults.** "within nothing," "one places," a duplicated milestone line.
- **Names.** "Rpgne," "Choose," "Redbull": V3's naming ladder on unnamed cars. Pre-dates V4; the oldest open item.

Fixes made and verified

All on the branch at **c423b8b**, each checked by replaying the 5 October bin through the fixed build, with the V3 suite (138) and the V4 suite (12) green.

FAULT	FIX	EVIDENCE ON THE REPLAY
Playback by name	Device names resolve to a PortAudio index at start-up, DirectSound preferred (it resamples; WASAPI rejects 24 kHz on a 48 kHz device). Match is exact or MME's 31-character truncation; never a shorter heuristic. Every failure logs its exception type and message, loudly, the first five times.	Resolution logged; indices in the manifest. Live confirmation still needed.
Camera on AI stories	A story moment takes the camera only when a human is in the story. AI-only stories are narrated and related, never shot.	Hold on humans 11% → 27%
Relate flood	A relate needs a story the human is behind or within two places and five seconds of, and no more than one relate across the broadcast every 30 s. AI-only contested positions and collapses below P3 no longer open at all.	Relates 64 → 8; lines 109 → 64
Lead-battle churn	Keyed on the leader; the chaser may change inside the story, announced once after a ten-second hold and not again inside 30 s.	Churn gone; "A new challenger" once
Model unused	Story kinds route to the model by default in hybrid mode.	Needs a live run with a key
Phrasing	Gaps under 0.15 s take a nose-to-tail variant; tenths are spoken as words; "one place"; milestone lines gated on direction.	Read on the replay

THE REPLAYED SCRIPT, SAME BIN, FIXED BUILD

64 lines · 55 story lines · 8 relates · 93% of story lines with a human subject or anchor · longest silence 44 s. The lead battle at the restart now reads: *"Redbull is all over the back of Ronin. The lead is on." ... "Redbull is right on the back of Ronin. Nose to tail for the lead." ... "Settled at the front for now. Ronin ahead of Redbull by 8.2 seconds."*

Recommendations

- 1. Make the audio path a product, before any more story work.** The device bug has now cost three evenings on two machines. Build the rest of the 2 October design: ship with nothing pinned (Windows default output works with no setup), a numbered first-run picker saved per user outside the repo, a start-up sound check that plays a tone down each leg and confirms frames were delivered, and an `--audio-check` command so anyone can prove the pipes without running a race. `monitor_active` becomes true only when sound came out.
- 2. Run the replay gate before the next live run.** Twenty minutes on the OK PC or Mike's rig: identity gate (`--stories off` against the Pass 4 expected set), then `s11` and `s05` with stories on. The 5 October bin now joins that corpus as the first league capture with the story layer, and today's fixes were tuned on it, so the gate also guards against over-fitting to one race.
- 3. Read the cuts, not just the lines.** 27% hold time on two humans is still low. The cuts file says which layer held the camera and why; the next tuning target is the V3 protected moments (lead change, AI collision at the front) and the away-max return, not the story layer.
- 4. One live run, template-only, speech on, on Mike's rig,** with OBS monitoring on as the second route, before the model is switched back in. Settle the audio question with a second pair of ears, then add the model as the next single variable.
- 5. Fix naming before the next league night.** Five AI names in this script were placeholders. The uniqueness rule and position fallback from the 25 September findings are one function and one assertion, and they touch every line in every mode.
- 6. Keep the bin and the OBS recording.** Together with the replayed script they are the first before-and-after pair this project has: what aired, and what the same race sounds like through the fixed build.

DECISIONS RECORDED

- Device strings: exact match or MME truncation only; a shorter configured name never matches a longer device (the "Speakers" rule stands).
- Story moments are human-only for the camera. AI stories get voice and relation, not the shot.
- A relate is a line only when the story can affect the human; proximity alone earns nothing.
- The hybrid writer treats story kinds as model kinds unless configured otherwise.